Grounded Two-Way Visual Dialogue: Interactive Multimodal Reasoning, Cross-View Geometric Verification, and Calibrated Perception

PhD Project Description & Research Proposal

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1. Vision: Beyond Text-Only Interaction to Two-Way Visual Dialogue

Contemporary Multimodal Large Language Models (MLLMs) have achieved remarkable conversational fluency, yet human–AI collaboration in the physical world demands going **beyond text-based interaction**. In real-world environments (e.g., robotics, collaborative design, augmented reality, medical inspection), humans and intelligent agents communicate through continuous visual references, pointing, spatial boundaries, and reciprocal visual feedback.

A truly intelligent visual partner must possess **two-way visual dialogue capabilities**:

1. **Input Comprehension:** Accurately parsing dense, ambiguous human visual cues (e.g., "the second tool behind the monitor", fine-grained masks, multi-view camera perspectives).
2. **Reciprocal Visual Generation:** Responding not merely with natural language descriptions, but with grounded visual actions—generating dynamic segmentation masks, 3D bounding boxes, spatial trajectories, and clarifying visual queries when ambiguous.
3. **Calibrated Spatial Confidence:** Honest estimation of visual uncertainty, knowing when to ask for clarification rather than hallucinating spatial locations.

This research proposal outlines a 3-year research plan within the **Visual Computing Section at DTU Compute** to formulate, train, and evaluate foundational architectures for grounded two-way visual dialogue.

2. Methodological Foundation & Prior Research

My previous research establishes the core building blocks for this agenda:

- **Label-Free Geometric Process Supervision (*Gah et al., SACAIR 2026*):** Supervised multimodal reasoning in **Qwen2.5-VL** by projecting 2D open-vocabulary predictions (**SAM3**) into shared 3D coordinates (**RegionPLC, OpenScene**), utilizing physical cross-view geometric consistency as an intrinsic reward for online **Group Relative Policy Optimization (GRPO)** in **VERL**.
- **Anti-Reward-Hacking & Validate-Before-Train Protocols:** Discovered that spatial grounding policies exploit recall-heavy metrics by predicting oversized bounding boxes, and resolved this by mathematically re-anchoring supervision to count-based precision and weighted $F_{0.5}$ metrics ($\rho > 0.72 \rightarrow 0.334$).
- **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA) (*IEEE TPAMI Under Review*):** Formulated a probabilistic framework utilizing Platt confidence calibration and Bernoulli uncertainty estimation to coordinate frozen foundation models at inference time ($\Delta\theta = 0$), achieving SOTA on PASCAL-5^i (77.9% HM) and ScanNet200 (33.68% HM).

3. Proposed 3-Year Research Plan (DTU Compute)

Year 1: Interactive Multi-Turn Grounding & Disambiguation Datasets

- **Objective:** Build high-quality multi-turn visual dialogue benchmarks capturing sequential human–AI disambiguation (e.g., user pointing $\rightarrow$ agent proposes candidate masks with confidence scores $\rightarrow$ user refines $\rightarrow$ agent confirms).

- **Methodology:** Combine synthetic multi-view 3D environments (ScanNet++, Habitat) with human-in-the-loop interactive sessions to curate diverse visual reference trajectories.
- **Milestone 1:** Submission to **CVPR / ICCV** on multi-turn grounded visual dialogue benchmarks.

Year 2: End-to-End Two-Way Visual Dialogue Architectures with RLVR

- **Objective:** Develop MLLM architectures natively outputting interleaved text and spatial tokens (masks, 3D boxes, point coordinates) trained via Reinforcement Learning with Verifiable Rewards (RLVR).
- **Methodology:** Generalize our cross-view geometric consistency rewards to dynamic conversational turns, enforcing spatial temporal consistency across conversational steps.
- **Milestone 2:** Publication at **ECCV / NeurIPS** on end-to-end grounded conversational vision models.

Year 3: Uncertainty Calibration, Real-World Interaction, and International Exchange

- **Objective:** Integrate RCASA probabilistic calibration to enable real-time risk-aware visual dialogue and conduct the mandatory DTU international research stay.
- **Methodology:** Deploy models in interactive human–AI user studies (evaluating task completion time, cognitive load, and visual grounding accuracy). Complete PhD thesis writing and defense.
- **Milestone 3:** Major journal submission (*IEEE TPAMI / IJCV*) and DTU PhD Dissertation Defense.

4. Synergy with DTU Compute & Teaching Readiness

- **Research Environment:** DTU Compute's Visual Computing section provides the ideal world-class ecosystem for geometric computer vision, 3D scene understanding, and interactive human-centered AI.
- **Teaching Commitment:** Eager to contribute actively as a Teaching Assistant (TA) in DTU courses such as *Computer Vision*, *Deep Learning*, and *Machine Learning*, drawing on 4+ years of prior university lecturing experience.